TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Analysis of Group II (Pb, Cu, Cd, Hg, Sb, Sn)

The acid-sulphide group — reading the precipitate's colour first, then the separations and confirmations for lead, copper, cadmium, mercury, antimony and tin.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 6 · LEVEL 1 TERM 1

1. Aim and first evidence: colour

Aim: to analyse the Group-II precipitate (sulphides thrown down by H_2S from the acid filtrate, Experiment 4) and identify the cations present. Group II is the course's most colourful group, and colour is genuine evidence here:

SULPHIDE	COLOUR
PbS, CuS, HgS	Black (HgS sometimes red in its cinnabar form, black as precipitated)
CdS	Yellow
Sb_2S_3	Orange
SnS / SnS_2	Brown / yellow

A yellow precipitate in this group is therefore nearly a signature (Cd), an orange one points to antimony, and black precipitates need the wet separations of §2. Record the colour *before* any reagent touches the precipitate.

2. Outline separation

The group divides by a classic solvent: **yellow ammonium sulphide** dissolves the sulphides of Sb and Sn (forming thio-salts) while PbS, CuS, CdS and HgS remain — the "copper division" stays as residue, the "arsenic/antimony/tin division" goes into solution, from which acidifying reprecipitates Sb_2S_3 (orange) and SnS_2 (yellow) for confirmation. Where the laboratory course treats the two divisions separately, this sheet's confirmations below cover the members named in the syllabus list.

3. Confirmatory tests

CATION	SEPARATION HANDLE AND CONFIRMATORY TEST
Pb^{2+}	Dissolve the sulphide fraction in dilute HNO_3 (PbS dissolves, HgS does not); the solution with dilute H_2SO_4 gives white $PbSO_4$; or K_2CrO_4 gives yellow $PbCrO_4$; KI gives yellow PbI_2.
Cu^{2+}	The nitric-acid solution is blue; NH_4OH in excess → deep blue $[Cu(NH_3)_4]^{2+}$; potassium ferrocyanide → chocolate-brown precipitate.
Cd^{2+}	Yellow CdS itself, reprecipitated from a solution in dilute acid by H_2S (after Cu is removed — e.g. Cu masked/removed per laboratory direction — CdS falls yellow from the faintly acid solution).
Hg^{2+}	HgS is insoluble in dilute HNO_3 but dissolves in aqua regia; the diluted solution gives: with $SnCl_2$ — white precipitate turning grey (a famous, sensitive test); with NaOH — yellow HgO; with KI (dropwise) — scarlet HgI_2 dissolving in excess.
Sb^{3+}	Orange Sb_2S_3 dissolves in concentrated HCl; the solution diluted gives white oxychloride turbidity (SbOCl), clearing with tartaric acid; H_2S reprecipitates orange.
Sn^{2+}/Sn^{4+}	SnS (brown) dissolves in yellow ammonium sulphide and in concentrated HCl; the HCl solution reduces $HgCl_2$ — white, then grey precipitate (stannous present/formed). SnS_2 (yellow) dissolves in concentrated HCl similarly.

4. Notes

- Dissolving sulphides in HNO_3 oxidises sulphide to sulphur (milky/yellow specks in the solution are normal) — do not mistake separated sulphur for a new precipitate.
- HgS's refusal to dissolve in dilute HNO_3 is itself the first separation inside the black trio: filter, and Pb/Cu proceed in solution while mercury waits for aqua regia.
- All Group-II work with H_2S and aqua regia belongs in the fume cupboard; mercury residues follow the Experiment 5 waste rule.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A Group-II precipitate is bright yellow. Name the sulphide and cation, and one confirmatory step.
2. An orange Group-II precipitate dissolves in concentrated HCl and the diluted solution turns turbid white. Name the cation and the white compound (in words or formula).
3. How are PbS and HgS separated, using one reagent's refusal?
4. State the $SnCl_2$ test for mercury — both stages of the precipitate's colour.

5. What does yellow ammonium sulphide separate within Group II, and on acidifying that extract, which two sulphides return and in which colours?
6. A black Group-II precipitate's nitric-acid solution turns deep blue with excess NH_4OH. Name the cation and the complex responsible (Chemistry-I link).

Answer key

1. CdS — cadmium. Confirmation: dissolve in dilute acid and reprecipitate yellow CdS with H_2S from the faintly acid solution (with copper removed/masked as directed).
2. Antimony (Sb^{3+}); the white turbidity is antimony oxychloride (SbOCl), formed on dilution and clearing with tartaric acid.
3. Dilute HNO_3 dissolves PbS (and CuS) but not HgS: filter — lead proceeds in the filtrate (confirm as $PbSO_4$/$PbCrO_4$), mercury remains for aqua regia.
4. $HgCl_2$ solution + $SnCl_2$: first a white precipitate (Hg_2Cl_2), which with excess $SnCl_2$ turns grey, then nearly black (metallic mercury) — reduction in two visible stages.
5. It dissolves the antimony and tin sulphides as thio-salts, leaving PbS, CuS, CdS, HgS undissolved. Acidifying the extract reprecipitates Sb_2S_3 (orange) and SnS_2 (yellow).
6. Cu^{2+}; the deep blue is the tetraamminecopper(II) complex $[Cu(NH_3)_4]^{2+}$ — the coordination chemistry of Chemistry-I Topic 4.

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